LORAINE SANDOVAL ASCENCIO

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Education

UNIVERSITY OF CALIFORNIA, IRVINE — Irvine, CA

Doctor of Philosophy in Physics with Concentration in Astronomy and Astrophysics — June 2027

CALIFORNIA STATE UNIVERSITY, LOS ANGELES — Los Angeles, CA

Master of Science in Physics — May 2021

UNIVERSITY OF CALIFORNIA, LOS ANGELES — Los Angeles, CA

Bachelor of Science in Astrophysics — June 2019

Research Experience

GRADUATE STUDENT RESEARCHER — Irvine, CA

University of California, Irvine — June 2022 – Present
Advisor: Dr. Michael C. Cooper

- Plan and conduct observing runs with the Keck Cosmic Web Imager (KCWI), including selecting and observing targets for spectroscopic follow-up.
- Process raw observational data using the KCWI data reduction pipeline for low-surface brightness galaxies.
- Extract and analyze 1D spectra for 45 candidate ultra-diffuse galaxies, including measurements of redshifts and stellar-population properties.
- Analyze the star-formation histories, atomic gas content, and environments of ultra-diffuse galaxies to investigate the physical mechanisms driving their evolution and quenching.

GRADUATE STUDENT RESEARCHER — Irvine, CA

University of California, Irvine — Aug. 2021 – June 2022
Advisor: Dr. Asantha Cooray

- Simulated galaxy spectra with the Bagpipes modeling tool, incorporating the resolution of the upcoming SPHEREx NASA survey to assess observational capabilities.
- Investigated how variations in galaxy parameters (e.g., star formation history, metallicity) influence the spectral features SPHEREx will detect, informing survey design and analysis strategies.

INTERSTELLAR AND HELIOSPHERIC PHYSICS GROUP INTERN — Pasadena, CA

Internship at NASA's Jet Propulsion Laboratory — June 2020 – Aug. 2021

Advisors: Dr. Susan Terebey and Dr. Neal Turner

- Updated an existing code's (RadChemT) current model of gas dynamics in the protostellar disk of young stellar object L1527 for the molecule $C^{18}O$.
- Modeled and understood the physics of shocked gas from supersonic flow to determine how material enters the protostellar disk of young stellar objects and arrives at the protostar.

RISE MS-TO-PHD FELLOW — Los Angeles, CA

California State University, Los Angeles — June 2019 – Aug. 2021

Advisor: Dr. Susan Terebey

- Researched the methods used to model the gas dynamics and kinematics in protostellar disks.
- Innovated new dynamics to be incorporated into an existing model for the young stellar object L1527.

COSMOLOGY RESEARCHER Los Angeles, CA

University of California, Los Angeles Feb. 2019 – Aug. 2019

Advisor: Dr. Steven Furlanetto

- Developed a Python-based model to simulate ionizing radiation from O and B stars and predict recombination line emission in theoretical nebulae.
- Utilized the Cloudy photoionization software to analyze line intensity ratios (e.g., OII, OIII, SIII) and refine model parameters for emission nebulae.

ORIGIN OF THE UNIVERSE GROUP INTERN Pasadena, CA

Internship at NASA's Jet Propulsion Laboratory Sept. 2016 – Aug. 2017

Advisors: Dr. Graça Rocha & Dr. Krzysztof M. Gorski

- Extracted a temperature CMB angular power spectrum using 2015 Planck mission data and IDL algorithms available for conducting likelihood estimates.
- Replicated previously published results by Planck mission scientists in *Planck 2015 results XI. CMB power spectra, likelihoods, and robustness of parameters*.

MANUFACTURING ENGINEERING INTERN Pasadena, CA

Internship at NASA's Jet Propulsion Laboratory June 2016 – Aug. 2016

Advisors: Michael Verdin & Gustavo Maldonado

- Aided thorough research and compiled data on a list of electronic components, analyzing characteristics such as termination finish, hermeticity, and moisture sensitivity levels.
- Used 3D modeling tools to simulate a diode endcap and investigate solder fillet formation.

Publications

- **Sandoval Ascencio, L.**, Cooper, M. C., Zaritsky, D., et al. 2026, in prep.
  Gas-Rich Post-Starburst Ultra-Diffuse Galaxies: Quenching or Rejuvenation?
- Khim, D. J., Zaritsky, D., **Sandoval Ascencio, L.**, et al. 2026, *ApJ*, submitted
  SMUDGes Below the Surface: Clusters Confirm the Diversity of Low-Surface-Brightness Galaxies
- Zaritsky, D., Donnerstein, R., **Sandoval Ascencio, L.**, et al. 2026, *ApJ,* 1006, 48
  Systematically Measuring Ultradiffuse Galaxies. IX. A Gyr in the Life of Nearby Low Surface Brightness Galaxies
- **Sandoval Ascencio, L.**, Cooper, M. C., Zaritsky, D., et al. 2025, *OJAp*, 8, 110
  Caught in the Act of Quenching? — A Population of Post-Starburst Ultra-Diffuse Galaxies
- Terebey, S., **Sandoval Ascencio, L.**, Flores-Rivera, L., et al. 2025, *ApJ*, 990, 53
  The Dynamics of Infall and Accretion Shocks in the Outer Disk
- Khim, D. J., Zaritsky, D., **Sandoval Ascencio, L.**, et al. 2025, *ApJ*, 989, 154
  A Closer Look at an Unusual Ultradiffuse Galaxy

Teaching Experience

TEACHING ASSISTANT Irvine, CA

UCI June 2022 – Present

- Lead discussion sections for both lower and upper-division undergraduate level physics and astronomy courses.
- Design worksheets to succinctly review each week's lecture material.
- Host office hours to provide homework help by effectively communicating examples.

TEACHING ASSOCIATE (INSTRUCTOR, ONLINE) Irvine, CA

UCI June 2025 – July 2025

- Condensed a 10-week introductory astronomy curriculum into a 5-week accelerated online format while maintaining learning outcomes.
- Designed and recorded lecture videos (3+ per week) integrating interactive concept-check questions to promote active learning.
- Facilitated asynchronous discussions and provided timely feedback to support remote student engagement in a compressed schedule.

TEACHING ASSOCIATE (INSTRUCTOR, IN PERSON) Irvine, CA

UCI September 2024 – December 2024

- Crafted and delivered engaging introductory astronomy lectures three times per week for 80+ undergraduate students.
- Integrated polling activities into lectures to encourage critical thinking and assess real-time understanding of key concepts.
- Developed and implemented qualitative and quantitative assessments to measure student performance.
- Collaborated with the teaching assistant to design effective assignments aligned with course objectives.

SCIENCE TUTOR Los Angeles, CA

UCLA BruinCorps Oct. 2017 – June 2018

- Assisted 30+ high school and middle school students with classwork and homework in a classroom setting.
- Supervised 30+ students during work periods.
- In-class tutored by proactively checking in with students.

MATH & SCIENCE TUTOR Sylmar, CA

LAMC STEM Math Center Sept. 2014 – Aug. 2016

- Assisted 20+ college students in targeted subject areas in a group setting.
- In-class tutored during classroom exercises and proactively checked in with students.
- Facilitated workshops on selected topics in physics and math by effectively communicating examples.

Professional Service

GALFRESCA 2026 ORGANIZING COMMITTEE MEMBER Irvine, CA

UCI Sept 2026

- Co-organized the 2026 Galaxy Formation & Evolution in Southern California (GalFRESCA) conference, bringing together more than 60 researchers from institutions across Southern California.
- Coordinated conference programming and logistics, including website building, scheduling talks, and communicating with participants.

SACNAS NATIONAL LIAISON Irvine, CA

UCI Sept. 2026 – June 2027

- Co-advise the graduate and undergraduate boards and foster collaborations with chapters across campuses.
- Track chapter events and milestones for annual reporting to SACNAS National.

SACNAS GRADUATE PRESIDENT Irvine, CA

UCI Sept. 2025 – June 2026

- Led the Graduate Executive Board in supporting underrepresented students in STEM through community building, professional development, and outreach.

- Oversaw academic, social, and outreach programming and coordinated collaborations with campus and UC partners.
- Mentored undergraduate board members and strengthened partnerships with campus organizations.

SACNAS SOCIAL CHAIR Irvine, CA

UCI Sept. 2024 – June 2025

- Organized inclusive social and networking events to foster community among SACNAS members.
- Mentored undergraduate social coordinators and collaborated with campus organizations on joint events.

PHYSICS GRADUATE CAUCUS SOCIAL EVENTS CHAIR Irvine, CA

UCI Sept. 2022 – June 2024

- Organized academic and social events to foster community among physics graduate students.
- Increased student engagement through initiatives such as themed socials and stress-relief activities.

PHYSICS AND ASTRONOMY COMMUNITY EXCELLENCE (PACE) MENTOR Irvine, CA

UCI Sept. 2022 – June 2023

- Peer mentored a first-year graduate student in the Physics and Astronomy Department.
- Met biquarterly for in-person meetings to discuss future research plans and offer advice for finding an advisor.

RISING STAR GIRLS TEAM MEMBER Irvine, CA

UCI Summers 2022 – 2026

- Assisted in preparation for a future week-long workshop designed to encourage underrepresented middle school girls to explore the sciences.
- Led workshop activities as a facilitator in an effort to combine both art and astronomy.
- Translated documents (registration, media release, etc.) for Spanish-speaking parents or guardians.

TRANSFER STUDENT MENTOR Los Angeles, CA

UCLA Transfer Center Sept. 2018 – June 2019

- Guided new transfer students in their transition to UCLA by sharing knowledge based on previous experience.
- Acquired information for new transfer students by attending mentor workshops.
- Built community among a cohort of 5 students through regular meetings and communication.

PRESIDENT OF ASTRONOMY CLUB Los Angeles, CA

LA Mission College Astronomy Club Sept. 2016 – June 2017

- Planned stargazing events, guest speakers, and trips with the help of board members.
- Organized and led biweekly meetings.

Academic Conferences

- NSBP - NSHP Joint Conference, Nov. 2026
- NDiSTEM Conference, Oct. 2025, 2026
- Keck Science Meeting, Sept. 2025, 2026
- Galaxy Formation & Evolution in Southern California (GalFRESCA) Conference, Sept. 2024, 2025, & 2026
- 10 Years of Ultra-Diffuse Galaxies: Past Perspectives and Future Prospects, Aug. 2025
- Sierra Conference for UC Astro Grads, Sept. 2024
- 236th Meeting of the American Astronomical Society (AAS), June 2020 (remote)
- Cal State LA Women in Data Science (WiDS) Conference, Mar. 2020
- 28th Annual Cal State LA Student Symposium on Research, Scholarship and Creative activities, Feb. 2020

Invited Talks and Community Engagement

- Lunch Seminar for the Department of Astrophysical and Planetary Science at CU Boulder, Nov. 2026
- Solar System Exhibit UCI Volunteer at the OC Discovery Cube, Mar. 2025 & 2026
- Science Demonstrations with SACNAS at the Downtown Santa Ana Artwalk, Dec. 2025
- SACNAS at UCI Transfer Visit Day Panel, May 2025
- Los Angeles Mission College STEM Alumni Panel, Mar. 2025
- Magnolia Public Schools STEAM Expo Judge, April 2024
- Magnolia Public Schools Science Fair Demonstration for Students, Mar. 2024
- Cal State LA Departmental Colloquium, Aug. 2020
- John F. Kennedy High School College Panel, Mar. 2019
- LA Mission College STEM Coffee Hour, Sept. 2018
- UCLA Women and Youth Supporting Each Other (WYSE) Empowerment Day Student Panel, Apr. 2018
- LA Mission College Society of Hispanic Professional Engineers (SHPE) Student Panel, Mar. 2017

Professional Development & Training

- CSU Pre-Professor Program (PREPP) Fellow, CSU Long Beach, Spring 2026
- Code/Astro software development workshop at Northwestern University, July 2023
- Observational Astronomy Workshop at UCO/Lick Observatory, May 2023

Certifications

- Summer Teaching Apprenticeship Program (STAP), Aug. 2025
- Mentoring Excellence Program (MEP), Feb. 2025
- Activate to Captivate Public Speaking, Mar. 2024
- Industry Insights for STEM Scientists, Nov. 2023

Fellowships & Scholarships

- José Flores Velázquez Endowment in Astrophysics, University of California, Irvine, June 2025
- Graduate Opportunity Fellowship, University of California, Irvine, 2021-2022
- Research Initiative for Scientific Enhancement MS-to-PhD Fellowship, CSU Los Angeles, Minority Opportunities in Research Programs, 2019-2021
- UCLA Regents Scholarship, 2017-2019

Honors & Awards

- UCI Latino Excellence and Achievements (LEAD) Award for Graduate Student Excellence, April 2026
- California State University, Los Angeles Special Recognition in Graduate Studies, May 2021
- UCLA Dean’s Honor List, Winter 2019 & Spring 2019
- UCLA Transfer Mentorship Program Certificate of Achievement, 2019
- LA Mission College President’s Honor List, Spring 2017
- LA Mission College Dean's Honor List, Spring 2017, Fall 2016, Spring 2016, Spring 2015, & Fall 2014

Skills & Abilities

DATA HANDLING

- Processing, analyzing, modeling, and managing large-scale astronomical datasets, including image reduction, spectral analysis, and multi-wavelength data integration

COMPUTER PROGRAMMING LANGUAGES

- Proficient in Python; experience with C, C++, and IDL

LANGUAGE

- Proficient in Spanish language